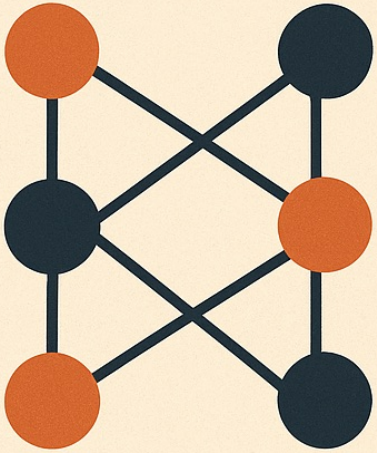
INTRO TO MACHINE LEARNING

AND NEURAL NETWORKS





Lecture 2: Linear Regression

#Agenda

1) Linear Regression

Obj ⟶ Final goal

(Dataset)

Theory + coding

Machine Learning

80%

Neural Networks

20%

1) ML & NN → Linear Reg & Logistic Reg

2) Supervised ML

3) Unsupervised ML

4) Neural Network → CV & NLP

ML prob Statement

Classification

Dog Vs Cat, Spam Vs Ham

Mango Vs Orange

Regression

predict the sales value for the nxt quarter

"House prices"

Clustering

Segment my customer to understand whom to give discount

Recommedation

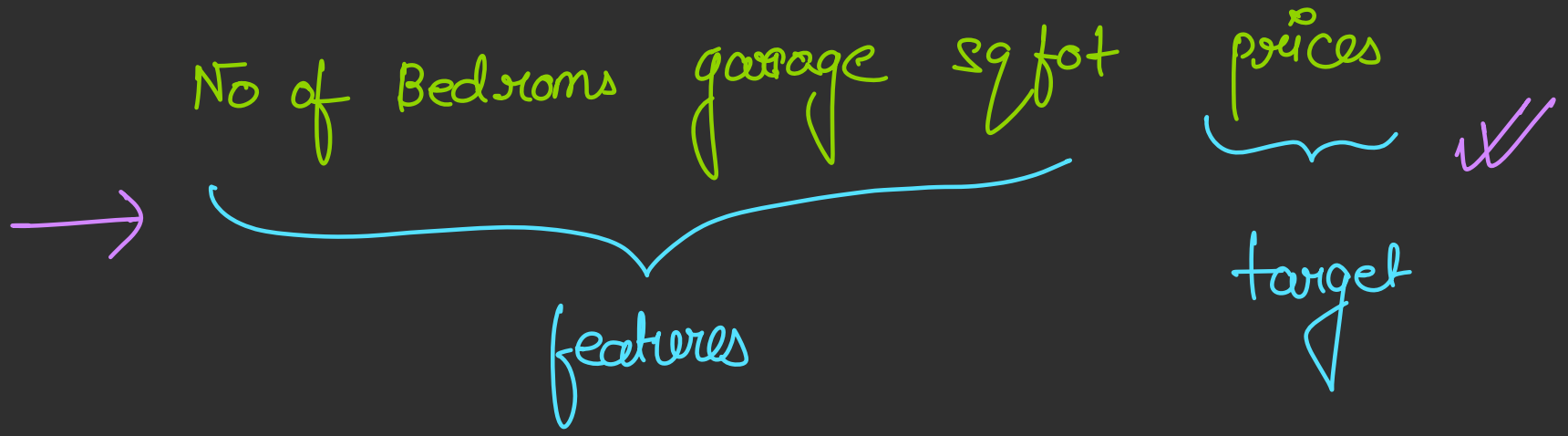
Youtube, Netfix

Forecasting

predict of stock prices for the next year

Supervised learning

Predict of House price



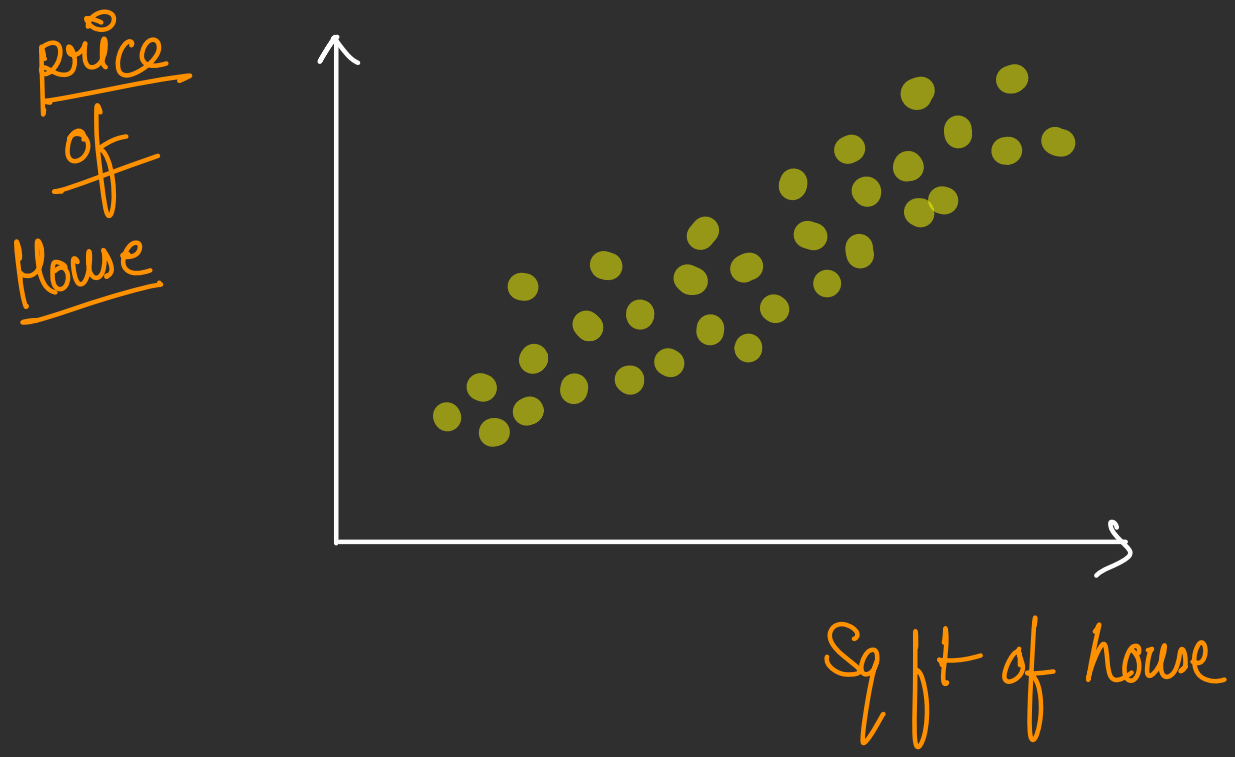
What is Regression Type of problem?

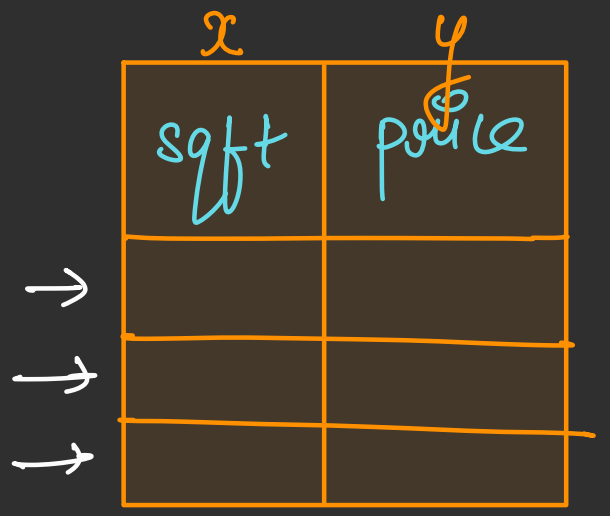
$x \longrightarrow$ features (categorical / numerical / dates / boolean)

$y \longrightarrow$ target (Numerical)

Price (target var)

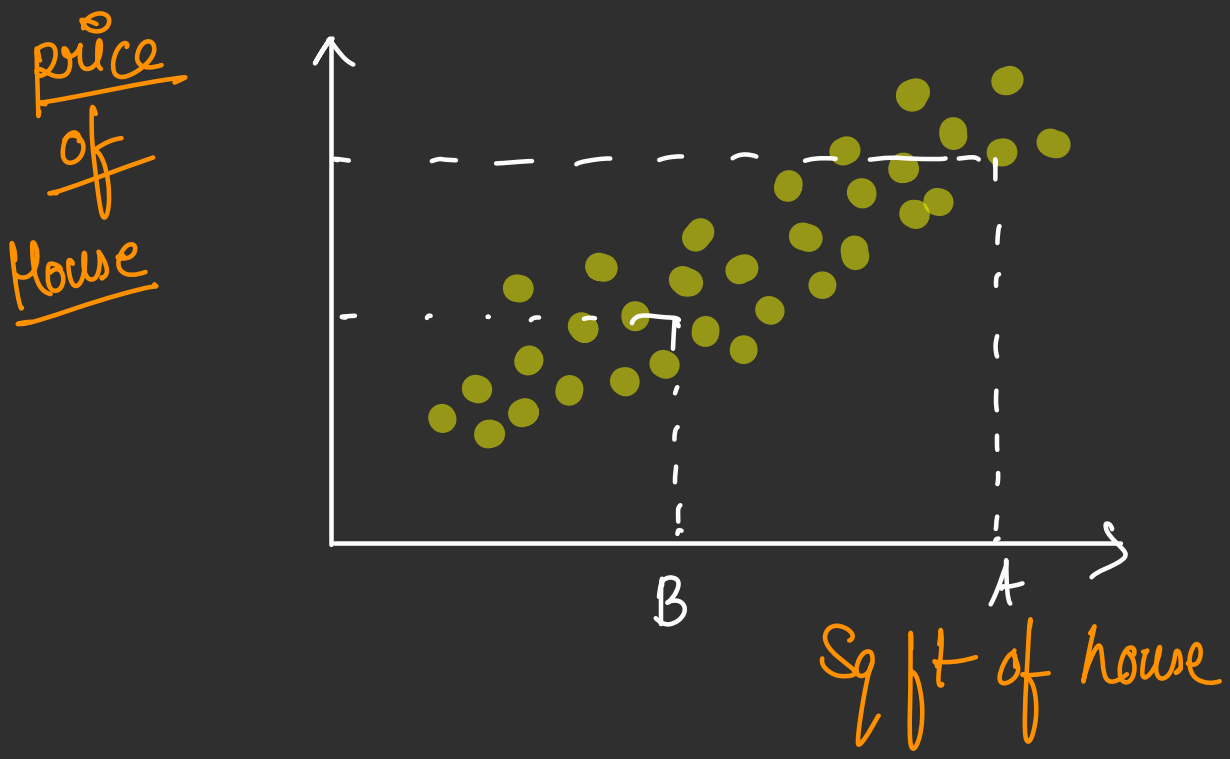
↳ Numerical





Regression

↳ It follows



Case Study

"Cars24" ⟶ Sells preowned cars

↳ DS @ Cars24

Make, Model . . . . .

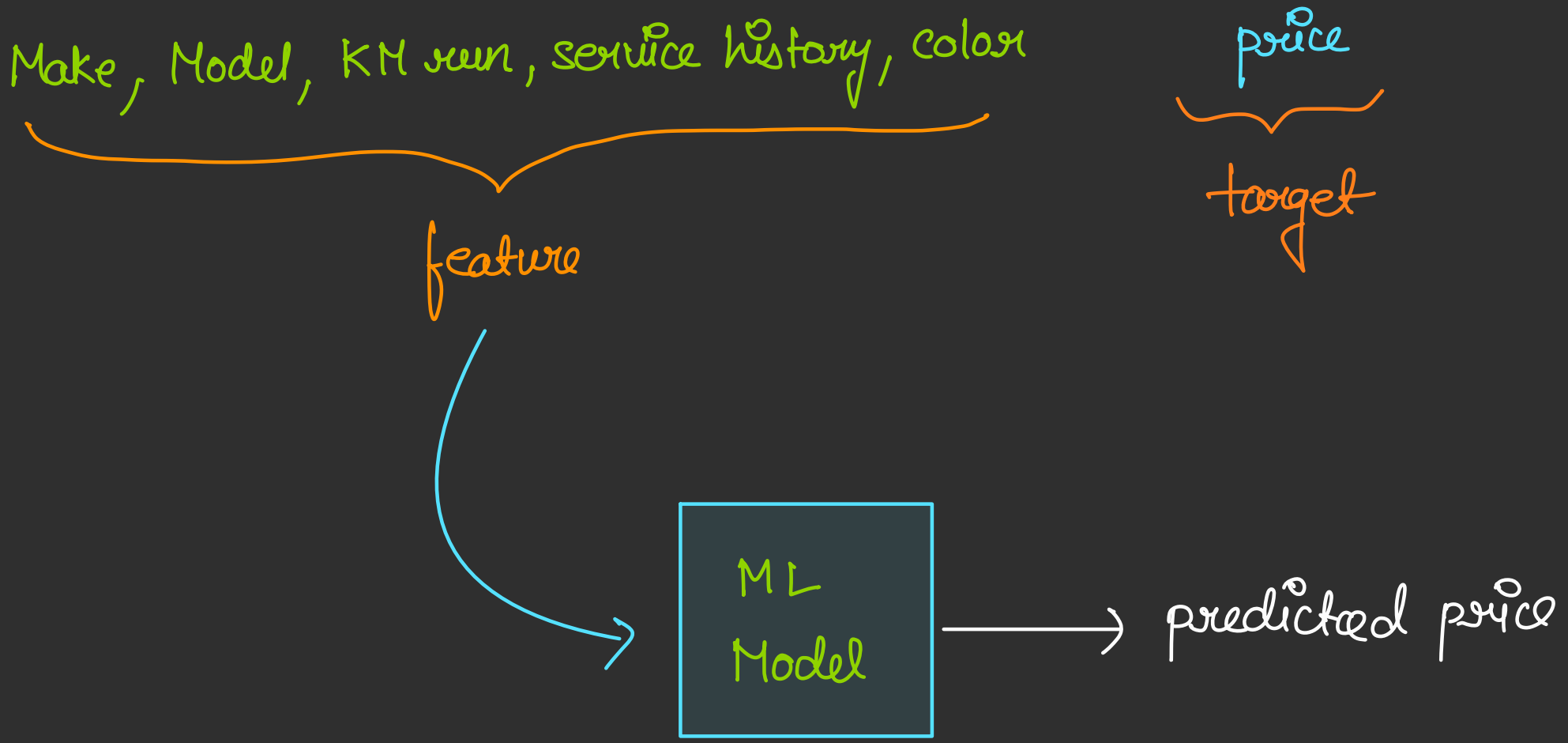
Could you predict the price of the car?

Price of the car

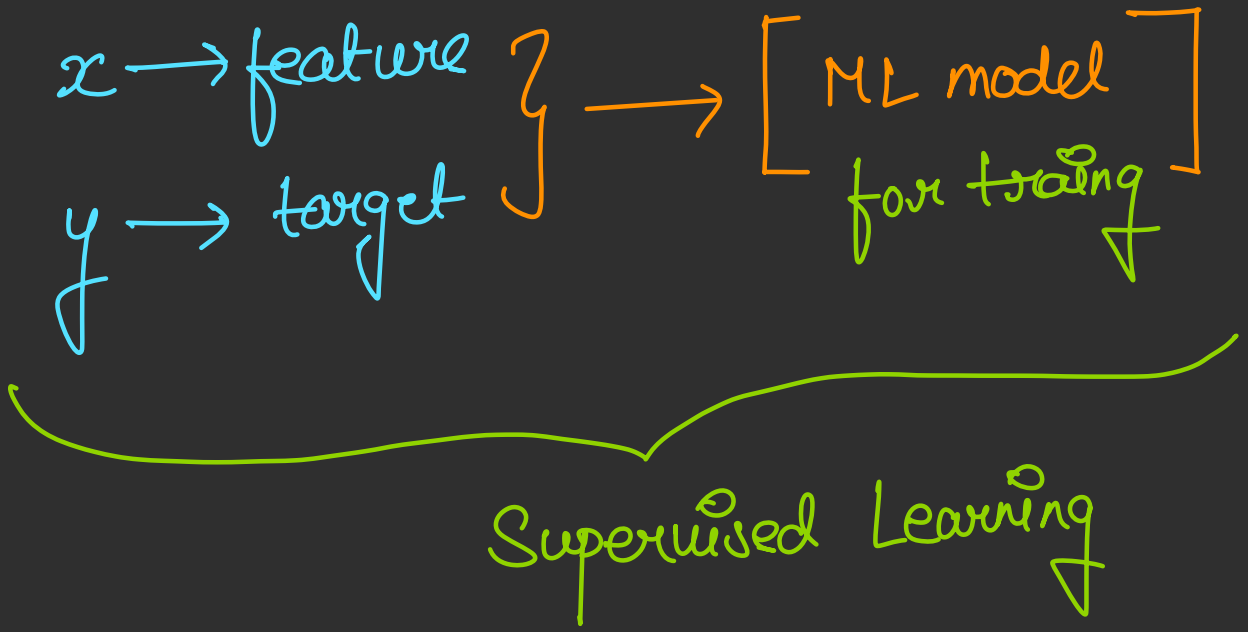
target

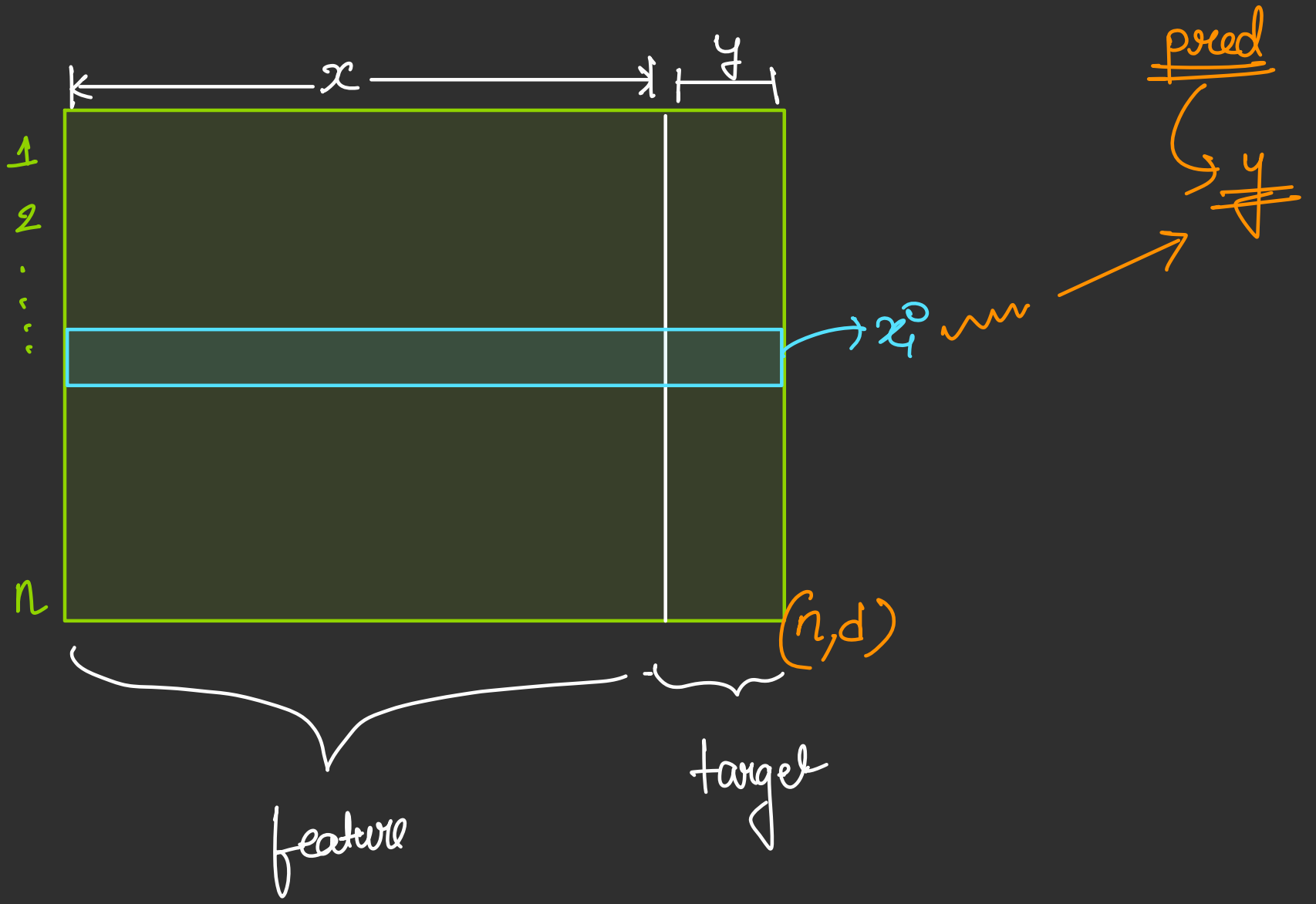
↓

Num ⟶ "Regression"


Make, Model, KM run, service history, color
price
target
feature
ML Model
predicted price

① Train the Model



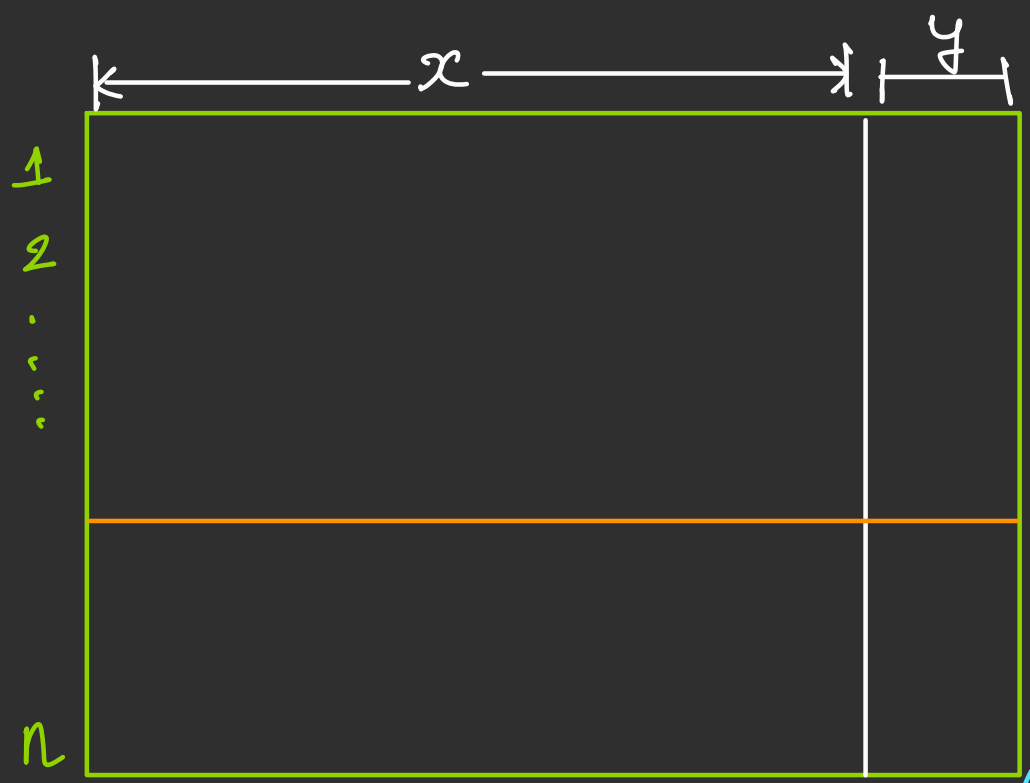

x
y
1
2
n
x_i^0
pred
$\neq y$
(n,d)
feature
target

y → Actual label

x_i → [Model] → $\hat{y}_i$

predicted value

$\hat{y} \simeq y$ { Good Training }



Entire data

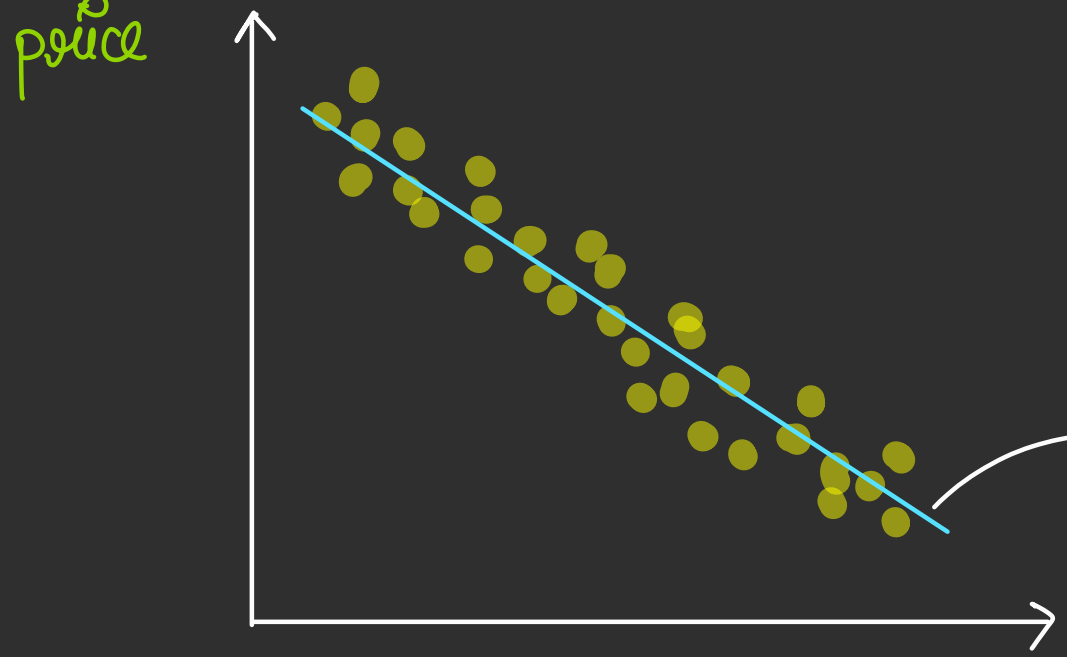
80% of my data "Training"

20% Testing

(n, d)

Linear Regression

goal: To find the best fit line that passes through almost every point.

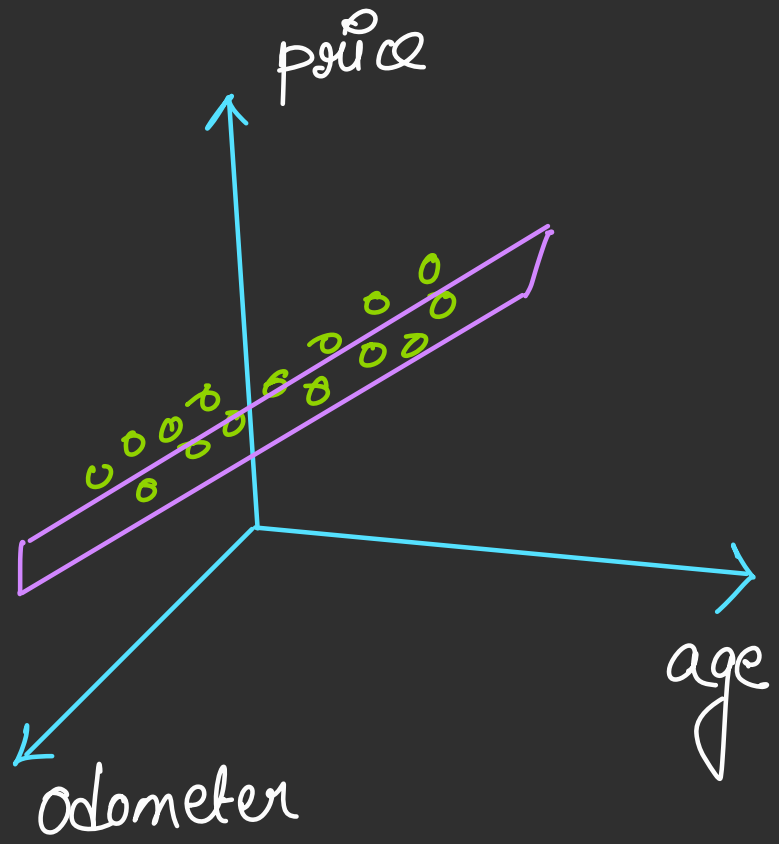


$y = mx + c$

- m → slope
- c → intercept

$y = w_1 x + w_0$

"Univariate data"



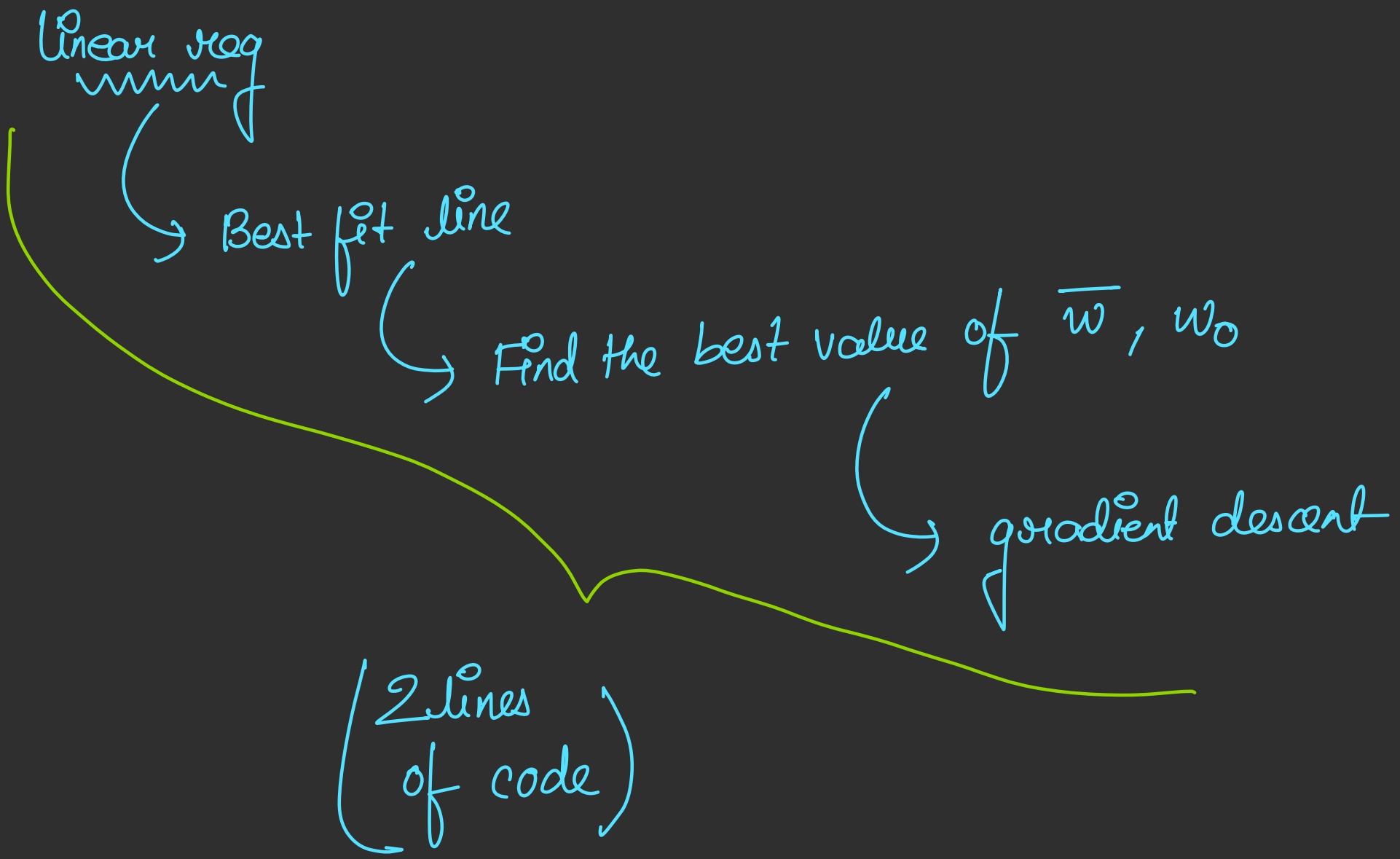
$$y = w_1 x_1 + w_2 x_2 + w_0$$

x_1: age

x_2: odometer

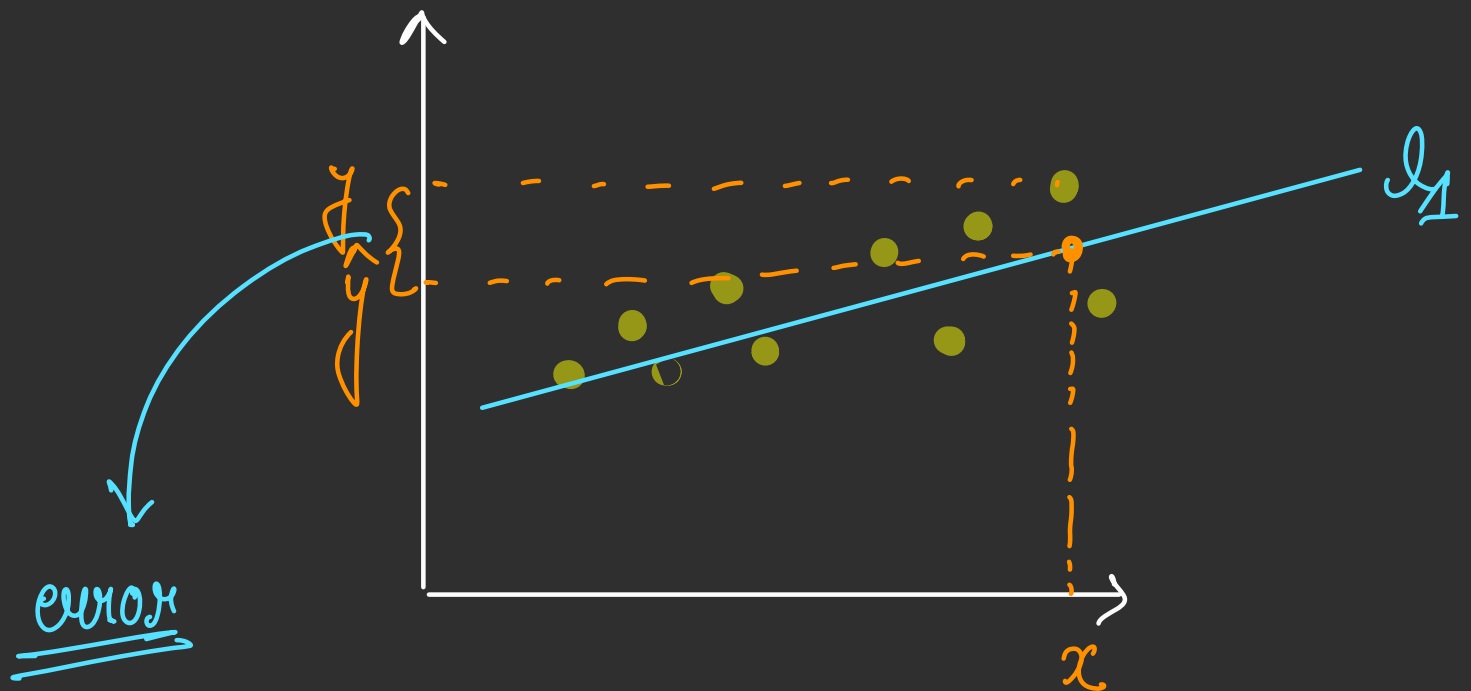
y: predicted price

$\hat{y}$


Linear reg
Best fit line
Find the best value of $\overline{w}$, w_0
gradient descent
(2 lines of code)

$$\hat{y} = w_1 x_1 + w_2 x_2 + \cdots + w_{17} x_{17} + w_0$$

Evaluation Metrics



$y_i - \hat{y}_i$ = error → minimize

$$\min_{w, w_0} \sum_{i=1}^{n} e_i$$

e_i

Loss Function

$$\min_{w, w_0} \sum_{i=1}^{n} \left(y_i - \underbrace{(w^T x_i + w_0)}_{\text{pred } (\hat{y})} \right)$$

1) -6

2) $+12$

3) -6

Mod

$$\frac{1}{m}\sum_{i=1}^{m}|\hat{y}_i - y_i|$$

MAE → Mean Absoulute Error

Square

$$\frac{1}{m}\sum_{i=1}^{m}(\hat{y}_i - y_i)^2$$

MSE :→ Mean Squared Error

Loss ↓ = Good Model

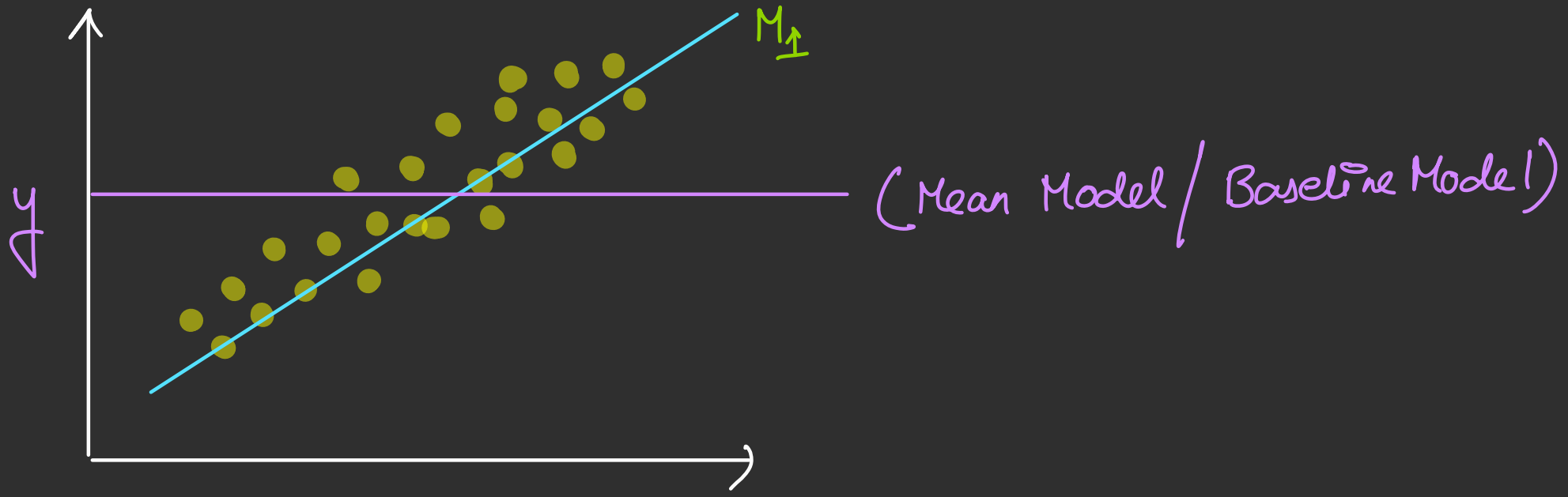
Data 1 $\neq$ Data

M_1 M_2

MSE: 21 MSE: 31

(You can only tell iff there is some baseline reference)

"R2 Score"



$R^2 \longrightarrow$ Coeff of determination

$$R^2 = 1 - \frac{SS_{res}}{SS_{total}}$$

$$R^2 = 1 - \frac{SS_{res}}{SS_{total}} = \frac{M_1}{B_m} = \frac{\frac{1}{m}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}{\frac{1}{m}\sum_{i=1}^{n}(\bar{y} - y_i)^2}$$

$\bar{y}$ → mean

$R^2 \Rightarrow$ 0 → Bad, 1 → good

$$R^2 \Rightarrow 1 - \frac{SS_{res} \uparrow}{SS_{total} \downarrow 0} \Rightarrow \infty \left\{ \text{Worst than Baseline} \right\}$$

$$R^2 \Rightarrow -\infty$$

$R^2 \Rightarrow 1 - \frac{SS_{res}}{SS_{total}} \to 0 \Rightarrow 1$ { Perfect Model }

$\hookrightarrow R^2 = 1$

$R^2 \Rightarrow 1 - \frac{SS_{res}}{SS_{total}} \to 1$

$\hookrightarrow R^2 = 0$

$SS_{res} = SS_{total}$

Same as baseline model

$$MSE = \frac{1}{m} \sum_{i=1}^{m} (\hat{y} - y)^2$$

$MSE \rightarrow 0$

$\hat{y} \rightarrow y$

No error

Doubts

target encoding

df

↳ index

df[col1]

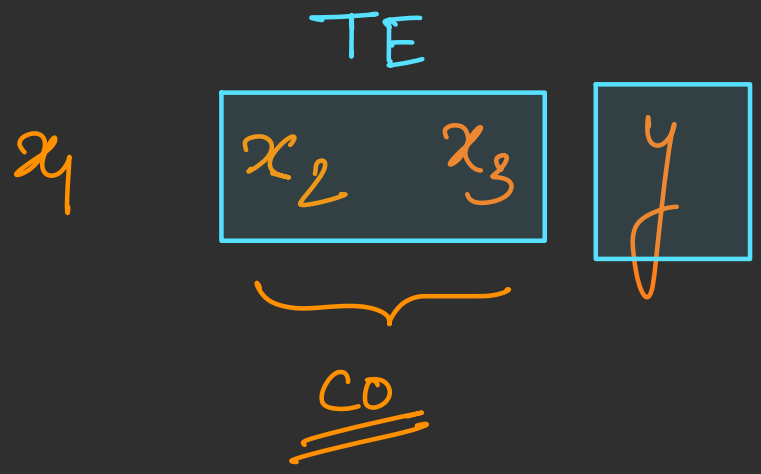
index

↳ Target Ecoble

Challenge

Target

[x ∝ y]



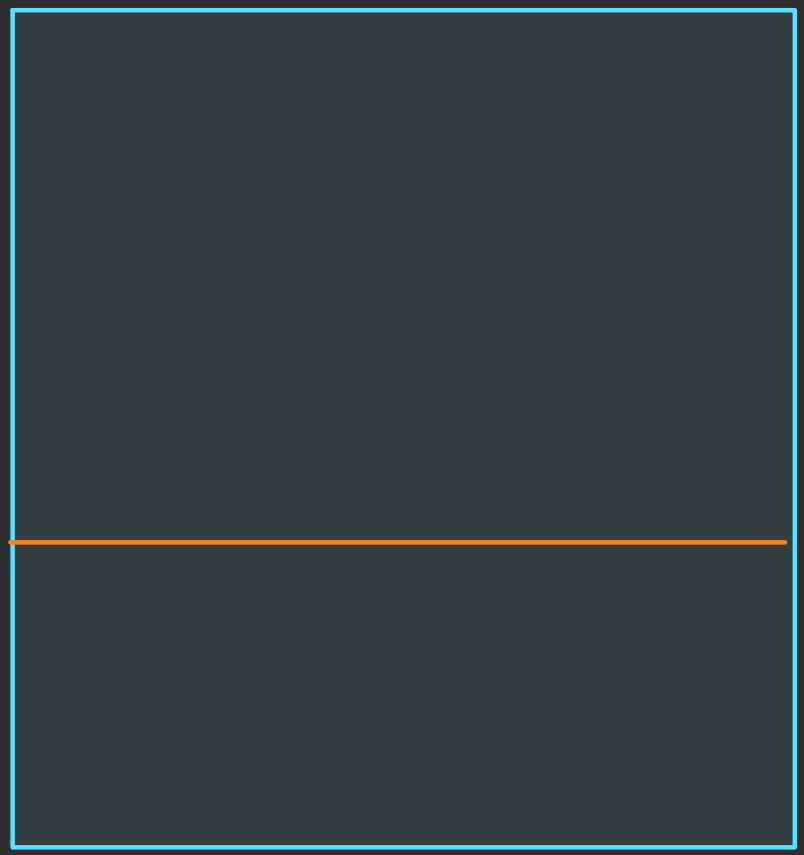
x_2 Vs y

x_3 Vs y

Corr is bad

Col

Happy Fox ⟶ 0.63



Training

Testing

70 : 20 : 10

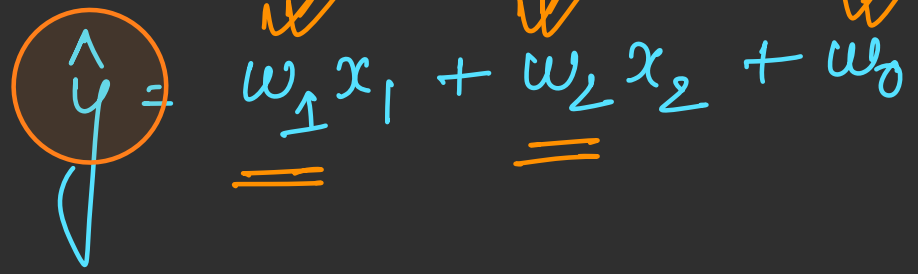
T

Tes

Va

Testing

Validation



MSE